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SOLVED NUMERICALS – JAN/FEB 2024 (CSE STREAM)

1(C) Estimate the attenuation in an optical fiber of length light 500m when a light signal of power 100mW emerges out of fiber with a power 90 mW.

(CSE Stream, JAN/FEB – 2024 | 22S)

Data: $Length\ of\ the\ optical\ fiber = 500m = o.5km$

$Input\ power\ of\ the\ signal\ , P_{in} = 100mW = 100 \times 10^{-3}W.$

$Output\ power\ of\ the\ signal\ , P_{out} = 90mW = 90 \times 10^{-3}W.$

To find: $Fiber\ attenuation, \alpha$=?

Solution: $The\ fiber\ attenuation\ \alpha\ is\ given\ by, \alpha = -\frac{10}{L}\ Log_{10}\left(\frac{P_{out}}{P_{in}}\right)$ dB/km

$$\alpha = -\frac{10}{L}\ Log_{10}\left(\frac{90\times10^{-3}}{100\times10^{-3}}\right) = 0.915\ dB/km$$

$$\therefore \boldsymbol{Fiber\ Attenuation = 0.915\ dB/km}$$

2(C) In a diffraction grating experiment, the laser light undergoes second order diffraction for diffraction angle 1.48º . The grating constant is 5.05×10^{-5}m and the distance between the grating and the source is 60cm. Find the wavelength of the laser light.

(CSE Stream, JAN/FEB – 2024 | 22S)

$Given\ that, m = 2, d = 5.05 \times 10^{-5}m, \theta = 1.48°$

$$Wavelength, \quad \lambda = \frac{d\ sin\theta}{m}$$

$$\lambda = \frac{5.05 \times 10^{-5} \times \sin 1.48°}{2}$$

$$\lambda = 6.5215 \times 10^{-7}m \quad Or\ \boldsymbol{6521.5 \times 10^{-10}m} \quad Or\ 6521.5A°$$

3(C) A particle of mass 0.5 MeV/c² has kinetic energy 100eV. Find its de-Broglie wavelength.

(CSE Stream, JAN/FEB – 2024 | 22S)

Data: Mass of the particle $m = 0.5$ MeV/c²,

Kinetic energy E = 100 x 1.609 x 10^{-19} J

Velocity of light, $c = 3 \times 10^8\ m/s$

Velocity of neutron $v = \frac{1}{10} \times c = 3 \times 10^7 m/s$

To find: De Broglie wavelength, λ = ?

Solution: De Broglie wavelength, $\lambda = \frac{h}{mv}$

We have, mass of the particle, m = $0.5\frac{\text{MeV}}{\text{c}^2} = \frac{0.5\times10^6\times1.602\times10^{-19}}{3\times10^8} = 8.9 \times 10^{-31}\ kg$

$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 8.9 \times 10^{-31} \times 100 \times 1.602 \times 10^{-19}}} = 1.24 \times 10^{-10}m$$

$$\therefore \boldsymbol{The\ De\ Broglie\ wavelength\ of\ the\ particle\ is\ 1.24 \times 10^{-10}m}$$

4(C) Calculate the energy of the first three states for an energy of an electron in one dimensional potential well of width $1A°$. (CSE Stream, JAN/FEB – 2024 | 22S)

Data: *Width of the potential well,* $a = 1Å = 10^{-10}m$

To find:

$Values\ of\ energy\ of\ the\ electron\ in\ the\ ground\ and\ the\ first\ two\ excited\ states$

$$i.e\ E_1 = ?, E_2 = ?, E_3 = ?$$

Solution: The energy of the electron in a one dimensional infinite potential well is given by,

$$E_n = \frac{n^2h^2}{8ma^2}$$

Where, the values of E that we get for *n=1, 2 and 3 correspond respectively to* E_1, E_2 *and* E_3.

$$\therefore E_1 = \frac{n^2h^2}{8ma^2} = \frac{1^2(6.63\times10^{-34})^2}{8(9.11\times10^{-31})(10^{-10})^2}$$
$$\therefore E_0 = E_1 = 6.0314\times10^{-18}J = 37.64\ eV$$

$$E_2 = \frac{n^2h^2}{8ma^2} = \frac{2^2(6.63\times10^{-34})^2}{8(9.11\times10^{-31})(10^{-10})^2} = 4\times6.0314\times10^{-18}\ J$$
$$\therefore E_2 = 2.4125\ \times10^{-17}J = 150.54\ eV$$

$$E_3 = \frac{n^2h^2}{8ma^2} = \frac{3^2(6.63\times10^{-34})^2}{8(9.11\times10^{-31})(10^{-10})^2} = 9\times6.0314\times10^{-18}\ J$$
$$\therefore E_3 = 5.4282\times10^{-17}J = 338.7\ eV$$

∴The values of Energy of the electron in the ground and the first two excited states are *respectively to* $\boldsymbol{E_1 = 6.0314\times10^{-18}J\ or\ 37.64\ eV}$,

$\boldsymbol{E_2 = 2.4125\ \times10^{-17}\ or\ 150.54\ eV\ and}$

$\boldsymbol{E_3 = 5.4282\times10^{-17}J\ or\ 338.7\ eV}$

5(C) ***A linear operator 'X' operates such that*** $\boldsymbol{X|0\rangle = |1\rangle}$ ***and*** $\boldsymbol{X|1\rangle = |0\rangle}$**. Find the matrix representation of 'X'. (CSE Stream, JAN/FEB – 2024 | 22S)**

Solution: $\begin{vmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{vmatrix}$, $|0\rangle = \begin{pmatrix}1\\0\end{pmatrix}$ *and* $|1\rangle = \begin{pmatrix}0\\1\end{pmatrix}$

$Given\ X|0\rangle = |1\rangle$ $\quad X|0\rangle = \begin{vmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{vmatrix}\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}0\\1\end{pmatrix}$

Multiply the matrices on LHS and equating with the matrix on the RHS we get, $x_{11} = 0$ & $x_{21} = 1$

$Given\ X|1\rangle = |0\rangle$ $\quad X|1\rangle = \begin{vmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{vmatrix}\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix}$

Multiply the matrices on LHS and equating with the matrix on the RHS we get, $x_{12} = 1$ & $x_{22} = 0$

$Therefore, X = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$

6(C) Explain matrix representation of 0 and 1 states and apply Identity operator $I|0>$ and $I|1>$ *states*.

(CSE Stream, JAN/FEB – 2024 | 22S)

$$We\ knoe\ that, Identity\ Matrix, I = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

$$I\,|0> = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}1\\ 0\end{bmatrix} = \begin{bmatrix}1\\ 0\end{bmatrix} = |0>$$

$$I\,|1> = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}0\\ 1\end{bmatrix} = \begin{bmatrix}0\\ 1\end{bmatrix} = |1>$$

Thus the operation of identity matrix(operator) on |0> and |1> leaves the states unchanged.

7(C) The superconducting transition temperature of Lead is 7.26 K. The initial field at 0 K is 50×10^3 Amp m^{-1}. Calculate the critical field at 6 K.

(CSE Stream, JAN/FEB – 2024 | 22S)

$$Given\ \ T_c = 7.26K, T = 6K\ and\ H_0 = 50 \times 10^3 Am^{-1}$$

$$H_C = H_0\left[1 - \left(\frac{T}{T_c}\right)^2\right]$$

$$H_C = 50 \times 10^3\left[1 - \left(\frac{6}{7.26}\right)^2\right]$$

$$H_C = 15849.32\ T\ Or\ 15.849 \times 10^3 Amp\ m^{-1}$$

8(C) Calculate the probability of an electron occupying an energy level 0.02eV above the Fermi level at 200K.

(CSE Stream, JAN/FEB – 2024 | 22S)

Data: $E - E_F = 0.02\text{eV} = 0.02 \times 1.602 \times 10^{-19} J$

To find: (i) $f(E)\ at\ 200K = ?$

Solution: $(i) Evaluation\ of\ \ f(E)\ at\ 200K = ?$

$$f(E) = \frac{1}{e^{\frac{E-E_F}{kT}}+1} = \frac{1}{e^{\frac{0.02\times1.602\times10^{-19}}{1.38\times10^{-23}\times200}}+1} = \frac{1}{e^{1.1594}+1} = \frac{1}{3.188+1} = \frac{1}{4.188} = 0.24$$

$$\therefore \boldsymbol{f(E)\ at\ 200K = 0.24}$$

$$\therefore \boldsymbol{Evaluation\ of\ \ f(E)\ at\ 200K = 0.24}$$

9(C) A slowing in object in an animation has a first frame distance 0.5m and first slow in frame 0.35m. Calculate the base distance and the number of frames in sequence.

(CSE Stream, JAN/FEB – 2023 | 22S)

$$base\ distance = \frac{distance\ between\ any\ two\ adjacent\ frames}{2}$$

$$base\ distance = \frac{0.5 - 0.35}{2} = 0.075m$$

$To\ find\ number\ of\ frames,$

$$No.of\ frames = \frac{total\ distance}{base\ distance}$$

$$No.of\ frames = \frac{0.5}{0.075} = 6.66\ \approx 7\ frames$$

9. (C) In an optical fiber experiment, the laser light propagating through optical fiber cable of length 1.5 m, made spot diameter of 8 mm on the screen. The distance between the end of the optical fiber and the screen is 0.031m. Calculate the angle of acceptance and numerical aperture of given fiber.

(CSE Stream, JAN/FEB – 2024 | 22S)

$$Given: D = 8mm\ or\ 0.8cm\ and\ l = 0.031m\ Or\ 3.1cm$$

$$Angle\ of\ Acceptance, tan\theta_0 = \left(\frac{D}{2l}\right)$$

$$Or\ \theta_0 = tan^{-1}\left(\frac{D}{2l}\right) = tan^{-1}\left(\frac{0.8}{2 \times 3.1}\right) = \mathbf{7.3523°\ 0r\ 7°21'}$$

$$Numerical\ Aperture, NA = sin\theta_0 = \sin(7°21') = \mathbf{0.1279}$$

10(C) While animating speeding up car animation, the total distance covered over 6 frames is 25m. Calculate the base distance.

(CSE Stream, JAN/FEB – 2024 | 22S)

$$Given: Total\ distance\ d = 25m\ and\ Frame\ number\ X = 6$$

$$Base\ distance, b = \frac{Total\ distance}{(Last\ frame - 1)^2} = \frac{d}{(X-1)^2}$$

$$b = \frac{25}{(6-1)^2} = \frac{25}{25} = 1m$$

$$\therefore Base\ distance, b = 1m$$